

December 2021

1 (a) Find the CRC for the data polynomial $x^9 + x^7 + x^5 + x^2 + x + 1$ with generator $x^3 + x + 1$.

Answer: Compute CRC by polynomial division. Append 3 zero bits (degree of generator) to the data bits and divide by 1011 (binary for $x^3 + x + 1$). Performing binary long division, the remainder (CRC) is **100** (i.e. x^2). In other words, the CRC bits are "100." This is the quotient remainder after dividing by the generator polynomial ¹.

1 (b) What is a LAN? What are typical characteristics of a LAN?

Answer: A **Local Area Network (LAN)** connects devices in a small geographic area (e.g. one building) ². LANs typically have **high bandwidth** (tens to hundreds of Mbps or more), **low latency**, and **low error rates**, using inexpensive media (like Ethernet cables or Wi-Fi). They operate over short distances (often under a few kilometers) and often use a single broadcast domain (all hosts can reach each other directly). Devices can be peer-to-peer or client-server. (For example, 100 Mbps or 1 Gbps Ethernet is common.) LANs also allow easy resource sharing (printers, files) within the network ².

1 (c) Compare transmission delay and propagation delay.

Answer:

- **Transmission delay (T_t)** is the time to push all packet bits onto the link. It depends on packet size and link data rate: $T_t = \frac{\text{Packet size (bits)}}{\text{Bandwidth (bps)}}$ ³. Larger packets or lower bandwidth give larger transmission delay.

- **Propagation delay (T_p)** is the time for a bit to travel through the medium. It depends on distance and signal speed: $T_p = \frac{\text{Link length (m)}}{\text{Signal speed (m/s)}}$ ⁴. For example, a 1000 km link at 2×10^8 m/s has $T_p = 5$ ms. Propagation delay is independent of packet size. In summary, transmission delay is "send-time" based on rate and size, while propagation delay is "travel-time" based on distance and medium ³ ⁴.

1 (d) Differentiate congestion control and flow control.

Answer:

- **Flow control** manages the sender's rate so the receiver's buffer is not overwhelmed. It is an end-to-end mechanism (often at data link or transport layer) that adapts transmission so the receiver can cope ⁵. For example, sliding-window protocols apply flow control.

- **Congestion control** manages the overall traffic in the network to prevent overload. It regulates the total sending rate in the network (typically at the transport layer) so intermediate routers do not become congested ⁵. For example, TCP's congestion window shrinks when the network is congested. In short, flow control is **receiver-centric** (avoiding receiver overflow) ⁵, while congestion control is **network-centric** (avoiding router queue overload).

1 (e) Compare a Layer 2 switch with a Layer 3 switch.

Answer:

- A **Layer-2 switch** operates at the Data Link layer. It forwards Ethernet frames based on MAC addresses and typically does **not** examine IP addresses ⁶. It essentially extends a LAN segment, keeping all devices in one broadcast domain. It builds a MAC address table by learning source addresses on each port. Layer-2 switches do **not** perform IP routing between subnets.

- A **Layer-3 switch** operates at the Network layer. It can route packets between IP subnets, using IP addresses and a routing table. It combines switching speed with basic routing functions. Layer-3 switches maintain IP routing tables and can interconnect LAN segments (VLANs) without a separate

router ⁶ . They support typical router features (static routing, possibly dynamic routing protocols). In summary, L2 switches forward by MAC addresses within one network, while L3 switches also forward by IP (perform routing) ⁶ .

1 (f) Explain RSA key generation algorithm with an example.

Answer: RSA is an asymmetric (public-key) algorithm. Key generation steps ⁷ ⁸ :

1. **Choose two distinct large primes** p and q .
2. Compute $n = p \times q$ and $\phi = (p - 1)(q - 1)$.
3. **Choose an encryption exponent** e such that $1 < e < \phi$ and $\gcd(e, \phi) = 1$.
4. **Compute decryption exponent** d as the modular inverse of $e \bmod \phi$ (so $de \equiv 1 \bmod \phi$).
5. Public key = (n, e) , Private key = (n, d) .

Example: Let $p = 7$, $q = 11$. Then $n = 77$, $\phi = (7 - 1)(11 - 1) = 60$. Choose $e = 17$ (which is relatively prime to 60). Compute d satisfying $17d \equiv 1 \pmod{60}$; one finds $d = 53$ (since $17 \times 53 = 901 = 1 + 15 \times 60$). Thus **Public key** = $(77, 17)$, **Private key** = $(77, 53)$.

1 (g) Discuss QAM modulation technique and draw an 8-QAM constellation diagram.

Answer: Quadrature Amplitude Modulation (QAM) uses two carrier waves 90° out of phase (in-phase and quadrature) and independently modulates their amplitudes ⁹ . Each symbol in QAM carries multiple bits by choosing one of several amplitude-phase combinations. 8-QAM has 8 distinct symbols (3 bits per symbol). For example, one 8-QAM constellation places 8 points in the I/Q plane (often symmetrically on a circle) each representing a 3-bit pattern.

Figure: Example 8-QAM constellation diagram (eight points on the I/Q plane).

In the diagram above, each point (marked "000", "001", ...) represents one of the 8 symbols (three bits). QAM increases spectral efficiency by encoding bits in both amplitude and phase. (Higher-order QAM (16-QAM, 64-QAM) use more points.) Here, 8-QAM might place points equally spaced in angle on a circle (as shown) or in an optimized pattern. The key is that each symbol corresponds to a unique combination of the two carrier amplitudes ⁹ .

1 (h) Draw the IPv4 header structure and explain the significance of the Fragment Offset field.

Answer: The IPv4 header is 20–60 bytes long, consisting (in order) of: Version (4b), IHL (4b), Type of Service/DSCP+ECN (8b), Total Length (16b); Identification (16b), Flags (3b) + Fragment Offset (13b); TTL (8b), Protocol (8b), Header Checksum (16b); Source IP (32b); Destination IP (32b); and optional Options. The **Fragment Offset** (13 bits) appears in the second word (with the Flags) and indicates where in the original data payload this fragment belongs. It is measured in 8-byte units. For example, if a packet was fragmented and the first fragment carried 400 bytes of data, the second fragment's offset would be $400/8 = 50$. The **Offset** allows the destination to reassemble fragments in the correct order. (The Flags bits include DF ("Don't Fragment") and MF ("More Fragments") ¹⁰ .) In summary, Fragment Offset = index of fragment's data relative to start of payload (each unit = 8 bytes) ¹¹ .

2 (a) Which frequency bands are used for AM, FM, and Radar? What is the relationship between tower height and repeater (cell) distance?

Answer:

- **AM Radio:** uses the Medium Frequency band. The AM broadcast band is roughly **530–1700 kHz** ¹² . (AM shortwave uses higher MF/HF.)
- **FM Radio:** uses the VHF band. The FM broadcast band is about **87.5–108.0 MHz** ¹³ .
- **Radar:** generally uses **microwave frequencies** (e.g. L-band, S-band, X-band, etc.). In practice, most radar operates in UHF/SHF ranges (hundreds of MHz to tens of GHz) ¹⁴ .

The line-of-sight (or radio horizon) distance increases with tower height. Approximately, the horizon distance d (km) is proportional to $\sqrt{\text{height (m)}}$. A rough formula is $d \approx 3.57\sqrt{h}$ (for height in meters) – taller towers mean repeaters can cover farther (common telecom rule-of-thumb).

2 (b) Why is PAM (Pulse Amplitude Modulation) needed as a prerequisite to PCM? What is the minimum sampling interval for a maximum frequency of 1 kHz?

Answer: PCM (Pulse Code Modulation) encodes analog amplitudes into digital values by first sampling the analog waveform. **PAM** is the intermediate step: it generates a train of pulses whose amplitudes follow the analog input. In other words, PCM usually samples the analog signal in time (via a sampler), producing a sequence of pulse amplitudes (PAM), and then quantizes those amplitudes to digital levels ¹⁵. Without PAM (pulse representation), you would not have discrete-time samples to quantize. Thus PAM is the sampling stage of PCM.

According to the sampling theorem, to sample a signal with highest frequency $f_{\max} = 1$ kHz without aliasing, the sampling frequency must be at least $2f_{\max} = 2$ kHz. The **minimum sampling interval** is therefore $T_s = 1/(2 \text{ kHz}) = 0.0005$ s (or **0.5 ms**).

2 (c) Explain circuit switching vs. packet switching with examples.

Answer:

- **Circuit Switching:** A dedicated path (circuit) is established between sender and receiver before communication, and remains reserved for the duration (e.g. a telephone call). No other user can use that reserved bandwidth. Example: Traditional telephone networks (or a dedicated leased-line) use circuit switching. Once connected, data (voice) flows continuously along the fixed route until teardown.
- **Packet Switching:** Data is broken into packets, and each packet is sent independently over the network. No dedicated path is reserved; packets from many sources share the links. Routers forward packets based on destination addresses. Example: The Internet (e.g. emails, web traffic) uses packet switching. Different packets may take different routes.

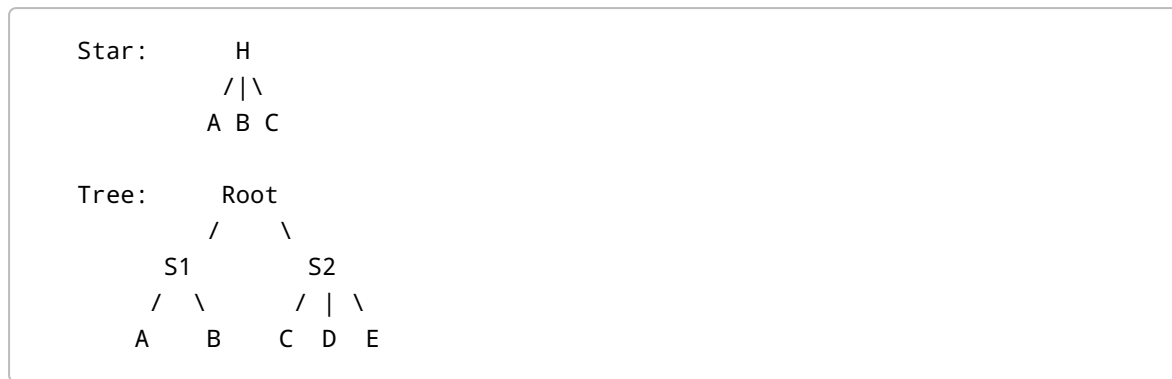
In summary, circuit switching has constant delay once set up but wastes resources during silence; packet switching is more flexible and efficient but requires buffering and allows variable delays. Packet switching also enables statistical multiplexing: many users share the same link dynamically.

2 (d) Compare star and tree topologies (with a diagram).

Answer:

- **Star Topology:** All hosts connect individually to a central node (hub or switch). (Diagram: draw a hub “H” in center with spokes to each node A, B, C.) Communication between any two nodes goes via the hub. If one link (node cable) fails, only that node is affected; if the central hub fails, the entire network is down. Star is simple to expand and troubleshoot.
- **Tree Topology:** A hierarchy of star networks. At the top is a root node; branches connect to secondary hubs, which in turn connect to more hosts (forming a tree structure). (Diagram: draw “Root” at top, branching to “Switch1” and “Switch2”; each switch connects to multiple end nodes.) This effectively combines multiple stars. Failure in a link disconnects the subtree, but others remain connected via other paths. Tree topology is scalable (just add branches) and suits segmented networks.

(See diagram below for conceptual illustration.)



Star: one central hub with every node directly attached. Tree: a hierarchy of hubs/switches (like branches) connecting groups of nodes.

3 (a) What is a Hamming code? How many redundant bits (parity bits) are needed for a 7-bit data word, and where are they inserted?

Answer: Hamming codes are error-correcting block codes that add parity bits to allow the receiver to detect and **correct** single-bit errors. The redundant parity bits are placed at positions that are powers of two (bit positions 1, 2, 4, 8, ...). For a 7-bit data word, determine the number of parity bits r by $2^r \geq m + r + 1$ (where $m = 7$). Testing, $r = 3$ gives $2^3 = 8 < 11$, $r = 4$ gives $2^4 = 16 \geq 11$, so **4 parity bits** are needed. These 4 parity bits are inserted at positions 1, 2, 4, and 8 of the 11-bit codeword ¹⁶. ¹⁷ The remaining positions (3,5,6,7,9,10,11) hold the original 7 data bits. Each parity bit covers certain data bits to enforce even (or odd) parity, enabling single-bit error correction.

3 (b) What is pure ALOHA? Calculate the throughput of Slotted ALOHA.

Answer:

- **Pure ALOHA:** A random access protocol where a station transmits whenever it has data, without waiting for any time slot. Collisions are likely, so after a collision each station waits a random time before retransmitting. There is no synchronization of transmissions.

- **Slotted ALOHA Throughput:** In Slotted Aloha, time is divided into discrete slots equal to one packet time, and stations may only begin sending at the start of a slot. The throughput S (in packets per slot) as a function of offered load G is $S = G e^{-G}$. The maximum throughput occurs at $G = 1$, giving $S_{\max} = 1/e \approx 0.368$ (36.8%) ¹⁸. In other words, even with optimal load, only about 0.37 successful packets are transmitted per time slot on average.

3 (c) Discuss IEEE 802.11 (Wi-Fi) protocol. Include a diagram if needed.

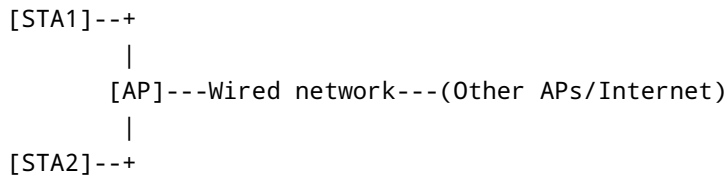
Answer: IEEE 802.11 is the standard for wireless LANs (Wi-Fi). It operates in the 2.4 GHz and 5 GHz bands (and newer 6 GHz, 60 GHz for 802.11ad/ay) and defines two sublayers at the MAC level:

- **DCF (Distributed Coordination Function):** The basic CSMA/CA protocol. Before sending, a station senses the channel; if idle, it may send immediately after a short **DIFS** wait. If the channel is busy, it backs off a random time. To avoid collisions (no CSMA/CD in wireless), it uses acknowledgments (ACKs): after sending a data frame, it waits for the ACK; if no ACK, it assumes a collision and retransmits.
- **PCF (Point Coordination Function):** An optional centralized polling mode, rarely used in practice, where an Access Point can poll stations for transmission.

802.11 framing includes MAC header fields (Frame Control, Duration/ID, Address1/2/3, Sequence Control, etc.) and supports management and control frames as well. In infrastructure mode, clients (STAs) connect through an Access Point (AP), forming a Basic Service Set (BSS). Multiple BSSs can be linked via a Distribution System (e.g. Ethernet) to form an Extended Service Set (ESS).

Key features: - **Carrier Sense / Collision Avoidance:** Because radios cannot detect collisions easily, stations use a backoff timer and can piggyback an ACK on a short interframe space (SIFS) after receiving a frame.

- **Frame types:** 802.11 defines management frames (beacon, association), control frames (ACK, RTS/CTS), and data frames. The MAC address fields carry source, destination, and BSSID (AP's MAC).



Above: An 802.11 *infrastructure network*: STAs communicate via the AP. The AP connects to other networks. Stations use CSMA/CA (802.11 DCF) to coordinate access, since wireless does not support collision detection ¹⁹ ²⁰. In summary, 802.11 provides shared-wireless access using CSMA/CA, supports multi-rate modulation (11b/g/n/ac/etc.), and allows mobile devices to connect over Wi-Fi.

4 (a) Explain the steps for connection-oriented transport connection establishment and graceful termination. Illustrate with a diagram.

Answer: The classic example is **TCP's three-way handshake** to establish a connection (and four-step shutdown). The steps are:

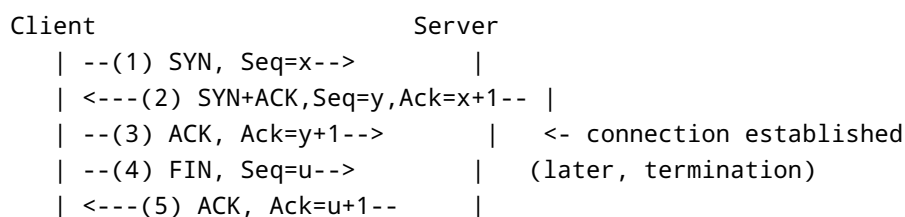
1. **Client -> Server:** SYN. The client sends a segment with the SYN flag set (and an initial sequence number).
2. **Server -> Client:** SYN+ACK. The server responds with its own SYN and acknowledges the client's SYN (ACK = client's Seq+1).
3. **Client -> Server:** ACK. The client sends an ACK to acknowledge the server's SYN (ACK = server's Seq+1).

After this exchange, the connection is established and data can flow bidirectionally.

To **terminate** (TCP tear-down), a four-way handshake is used: one side sends FIN; the other ACKs it and then sends its own FIN; finally the first side ACKs the second FIN. In detail (assuming client initiates close):

1. **Client -> Server:** FIN. (Close request from client.)
2. **Server -> Client:** ACK of client's FIN. (Client's side is now half-closed; client stops sending.)
3. **Server -> Client:** FIN. (Server's side closes, sending its FIN.)
4. **Client -> Server:** ACK of server's FIN.

After step 4, the connection is fully closed. Each FIN/ACK pair consumes one round trip (these steps are often drawn with arrows in sequence). The diagram is:



<---(6) FIN, Seq=v-->	
--(7) ACK, Ack=v+1-->	

*(1)-(3) is the 3-way connect; (4)-(7) is the 4-way close.) The key point is the three-phase handshake for setup (SYN, SYN+ACK, ACK) and four phases for tear-down, ensuring both sides agree on the connection teardown ²¹ ²² .

4 (b) Using Dijkstra's algorithm, find the best route (shortest path) between points A and E in the given network. (You must show your steps clearly.)

Answer: [General procedure given lack of specific graph]: Dijkstra's algorithm computes the shortest path by repeatedly selecting the unvisited node with the smallest tentative distance from the source, then relaxing its outgoing edges. The steps are: initialize distance of A = 0, all others = ∞ . Mark A visited. Update neighbors of A with their edge weights. Pick the unvisited node with smallest distance, mark it visited, and update its neighbors. Continue until E is reached. Due to no actual topology provided, we describe the method: at each step, write out tentative distances to each node, pick the smallest, and update. The final distance to E is the length of the shortest path, and tracking the predecessor pointers yields the route A→...→E.

For example, if the graph had edges $A-B = 3$, $B-E = 5$, $A-C = 2$, $C-E = 10$ etc., Dijkstra would find the minimal sum. (Without the specific graph values, we cannot compute numeric path; in an exam we would show a table of distances step-by-step.) The algorithm is summarized by: **choose smallest tentative distance node, update neighbors, repeat** ²³ .

5 (a) What is the difference between private and public IPv4 addresses? How are public IPs allocated?

Answer:

- **Private IP addresses** are special address ranges reserved by RFC 1918 for use within private networks. They are **not routable on the Internet**. The reserved private ranges are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16 ²⁴ . Devices on a LAN use these addresses internally; routers perform NAT so that all outbound traffic appears with a public IP.

- **Public IP addresses** are globally unique addresses assigned to devices that directly communicate over the Internet. These addresses come from the rest of the IPv4 space (outside the private ranges). Public IPs are allocated hierarchically: Internet Assigned Numbers Authority (IANA) distributes large blocks to Regional Internet Registries (RIRs), which allocate to ISPs. End users obtain a public IP either from their ISP or from their organization's block. In short, public IPs are assigned via IANA→RIR→ISP→end-user chain ²⁵ . Public IPs must be unique on the Internet; private IPs are reused in many networks.

5 (b) Explain the Diffie-Hellman key exchange concept and describe RSA key generation (with a small example).

Answer:

- **Diffie-Hellman (DH) Key Exchange:** Allows two parties with no shared secret to agree on a shared secret over an insecure channel. They agree publicly on a large prime p and base g . Alice picks a secret exponent a and sends $A = g^a \bmod p$; Bob picks b and sends $B = g^b \bmod p$. Each then computes $\text{Key} = B^a \bmod p = A^b \bmod p = g^{ab} \bmod p$. Even eavesdroppers knowing A, B, p, g cannot compute g^{ab} without solving the discrete log. In short, DH establishes a common secret $g^{ab} \bmod p$ from exchanged values ²⁶ .

- **RSA Key Generation:** (As in 1(f) above.) The steps are: choose primes p, q ; compute $n = pq$ and $\phi = (p-1)(q-1)$; pick public exponent e coprime with ϕ ; compute private exponent d with $de \equiv 1 \bmod \phi$ ⁸ . Example: $p = 7, q = 11 \implies n = 77, \phi = 60$. Choose $e = 17$ (since $\gcd(17, 60) = 1$). Compute d such that $17d \equiv 1 \pmod{60}$, which gives $d = 53$. Thus **Public key**=(77,17), **Private key**=(77,53) as before. RSA relies on the difficulty of factoring n .

June 2022

1 (a) What is the CRC? Write the polynomial representation of bitstring 1100010.

Answer: A CRC (Cyclic Redundancy Check) is an error-detection code created by treating bitstrings as polynomials and computing a remainder upon division by a generator polynomial ¹. The CRC is the remainder bits that ensure the whole message is divisible by the generator.

The bitstring **1100010** can be represented as a polynomial by assigning the leftmost bit to the highest power. If the bits are $b_6b_5 \dots b_0 = 1100010$, this corresponds to $x^6 + x^5 + x^1$ (since bits 6,5,1 are '1'). So $1100010 = x^6 + x^5 + x^1$.

1 (b) What is a wireless LAN? List disadvantages of using radio transmitters.

Answer: A Wireless LAN (WLAN) is a LAN that uses wireless (radio) links instead of cables. It allows devices (Wi-Fi stations) to connect to a local network via an Access Point (e.g. 802.11/Airport networks).

Disadvantages of radio (wireless) links include:

- **Interference and noise:** Radio signals are subject to interference from other devices or environmental factors, leading to variable performance.
- **Limited range:** Coverage is limited by transmitter power and obstructions. Walls and distance degrade signal.
- **Security vulnerabilities:** Wireless can be eavesdropped more easily, requiring encryption.
- **Shared medium:** All devices share the spectrum; bandwidth is typically lower than wired LAN for many users.
- **Multipath fading:** Signals reflect off surfaces, causing interference patterns.

While WLAN is convenient, these factors make it less reliable or more complex than a wired LAN.

1 (c) Explain transmission media. Compare optical fiber vs. copper.

Answer: Transmission media are the physical conduits that carry signals (electrical, optical, wireless) between devices. Two common wired media are optical fiber and copper (twisted-pair or coax).

- **Optical Fiber:** Carries information as pulses of light. It has extremely high bandwidth (gigabits and up), very low attenuation (suitable for long distances), and is immune to electromagnetic interference ²⁷. It can span many kilometers without repeaters. It is lightweight but can be expensive and requires special transceivers. For example, a single-mode fiber can transmit tens of Gbps over 10 km.
- **Copper (e.g. Ethernet cable):** Carries electrical signals. It has lower bandwidth (up to ~10 Gbps for Cat6A, higher speeds over short runs) and higher loss over distance. Copper is cheaper and easier to install but is susceptible to EMI and crosstalk ²⁸. It typically limits segment length (e.g. 100 m for Cat5e). Copper is convenient for short links (within buildings), while fiber is preferred for backbone and long-haul links.

In summary, fiber offers much greater capacity and range (and no EMI) ²⁷, whereas copper is less expensive and easier to handle but has lower performance and noise vulnerability ²⁸.

1 (d) What is a burst error? How can burst errors be corrected?

Answer: A **burst error** is a sequence of erroneous bits occurring in a row in the transmitted data. Formally, a burst error of length L means that L contiguous bits are in error, while the bits just before and after are correct ²⁹. For example, a sudden voltage spike might flip several consecutive bits (an "impulse" error).

Burst errors are corrected by using error-correcting codes or interleaving. For instance, **interleaving** spreads the data of one packet across multiple codewords so that a burst affects one bit in many codewords rather than many bits in one codeword. Then an error-correcting code (like a Reed–Solomon or convolutional code) on each codeword can correct the few isolated errors. Alternatively, a powerful block code (e.g. Reed-Solomon) designed to correct bursts can be used. In practice, combining error-correcting codes with interleaving is a common way to handle bursts.

1 (e) Describe three different types of addressing provided by the network layer. Give an example of each.

Answer: The network layer supports several addressing modes for packet delivery:

1. **Unicast:** One-to-one addressing. Each packet is delivered to a single, unique destination IP address. *Example:* Sending a web request from one PC to a specific server (destination IP 203.0.113.5).

2. **Broadcast:** One-to-all on a local network (only IPv4). The packet's destination is an all-ones broadcast address (e.g. 192.168.1.255 in a /24). All hosts on the local LAN receive it. *Example:* ARP requests use broadcast to find a MAC for an IP.

3. **Multicast:** One-to-many using special multicast group addresses (IPv4 224.0.0.0/4, IPv6 ff00::/8). The packet is delivered only to hosts that have joined the multicast group. *Example:* A streaming video multicast to multiple subscribers (IP 239.255.0.1).

Each addressing type serves a different purpose: unicast for normal one-to-one communication, broadcast for “everyone on the LAN,” and multicast for efficient group delivery.

1 (f) State three different service types provided by the transport layer to the application layer (with examples).

Answer: The transport layer can offer different service **semantics** to applications. Three examples are:

- **Unreliable datagram service:** No connection is established; data may be lost or arrive out of order (like UDP). *Example:* DNS queries use UDP (best-effort delivery with no guarantees).

- **Reliable connection-oriented stream:** A connection is established and the layer provides reliable, in-order delivery (like TCP). *Example:* HTTP over TCP ensures data arrives intact.

- **Multiplexing/demultiplexing and port addressing:** Transport provides multiplexing of multiple applications on one host via port numbers. Even the previous two services both rely on the transport demux. *Example:* A web server on port 80 and an email server on port 25 both use TCP/IP on one machine.

Other “services” include optional flow control and congestion control (TCP) or connection establishment (3-way handshake) for connection-oriented transport. In summary, typical transport services are **connectionless best-effort** (UDP), **connection-oriented reliable** (TCP), and the mechanisms of port-based multiplexing for both.

2 (a) Consider a 1-MHz bandwidth-limited channel. What is the Nyquist bit-rate for binary signalling? Also, what minimum sampling frequency is needed to sample a waveform whose highest frequency is 300 kHz?

Answer: If a channel has bandwidth $B = 1$ MHz and binary signalling (2 levels), the maximum bit rate by Nyquist's formula is $2B = 2 \times 1 \text{ MHz} = 2 \text{ Mbit/s}$. That is, the Nyquist rate for binary signals is **2 Mbps**.

For analog-to-digital sampling of a signal whose highest frequency $f_{\max} = 300$ kHz, the sampling theorem requires $f_s \geq 2f_{\max} = 600$ kHz. So the minimum sampling frequency is **600 kHz**, which implies a sampling interval of about $1/600000 \approx 1.67 \mu\text{s}$.

2 (b) What is the Nyquist criterion for two-level signalling over a band-limited channel? What is meant by baseband double-sideband transmission?

Answer: For two-level (binary) signalling over a noiseless bandwidth-limited channel, **Nyquist's criterion** states that the maximum symbol rate is $2B$ symbols/sec (where B is the channel bandwidth). In binary signalling, each symbol carries 1 bit, so the maximum bit rate is $2B$ bits/sec ³.

"Baseband double-sideband" transmission means the signal is transmitted without modulation onto a higher-frequency carrier (baseband). A double-sideband baseband signal has its spectrum symmetric around 0 Hz, extending from $-B$ to $+B$ if the baseband bandwidth is B . For example, an Ethernet cable carries baseband Manchester-coded bits; its spectrum has both positive and negative frequency components (double-sideband). (The term distinguishes from single-sideband or passband modulation.)

2 (c) What is an FTP Passive mode? Explain.

Answer: FTP (File Transfer Protocol) can operate in **active** or **passive** mode regarding data connections. In **passive mode**, the FTP server opens a random TCP port and waits; the client then connects to that port for data transfer. Specifically, the client sends the PASV command. The server replies with "entering Passive Mode (h1,h2,h3,h4,p1,p2)", encoding an IP and port. The client then initiates the TCP connection to that server-specified port. This is used when the client is behind a firewall/NAT and cannot accept incoming connections. Passive mode avoids the server trying to open a connection back to the client (which might be blocked), by having the client actively connect to the server's data port.

2 (d) Show how 16-bit two's-complement binary addition works, e.g. add 0101 1100 0011 0101 and 1011 0011 0010 1100.

Answer: Perform bitwise addition with carry (on 16 bits). For example:

```
0101 1100 0011 0101
+ 1011 0011 0010 1100
= 1 0000 1111 0101 1001   (with carry out of the 16th bit)
```

Ignoring overflow beyond 16 bits, the 16-bit result is **0000 1111 0101 1001**. If interpreting as two's-complement, one operand was positive ($0x5C35 = +23541$) and one negative ($0xB32C = -20276$), the result is $+3265$ ($0x0F59$). This manual addition uses standard binary addition rules (bitwise sum with carry).

3 (a) Why are virtual-circuit networks called "connection-oriented"? Why are datagram networks called "connectionless"?

Answer:

- A **virtual-circuit** (VC) network establishes a predetermined path (like a logical connection) through the network before data is sent. This mimics a persistent connection: all packets for that session follow the same path (in-order) once the VC is set up. Hence the service is **connection-oriented** (a route is reserved for the session). Examples: ATM or Frame Relay can use virtual circuits. The sender and receiver experience it like a continuous "circuit" (though it may be time-multiplexed).

- A **datagram** network does not set up any such path. Each packet carries full destination addressing and is routed independently. There is no prearranged connection state. Thus it is **connectionless**: packets may take different routes and arrive out of order or not at all. IPv4 over the Internet is connectionless (no VC is set up). Each router treats each packet separately.

In short, VC = path setup ("virtual circuit" like a phone call), whereas datagram = no setup, each packet "finds its own way" (like mailing letters).

3 (b) List the layers of TCP/IP model and map them to the OSI layers.

Answer: The TCP/IP model has 4 (or 5) layers that correspond to the 7 OSI layers:

- **Application (TCP/IP)** ↔ OSI Application, Presentation, and Session layers. (TCP/IP lumps all high-level protocols here: HTTP, FTP, SMTP, etc.)
- **Transport (TCP/IP)** ↔ OSI Transport layer. (TCP or UDP.)
- **Internet (TCP/IP)** ↔ OSI Network layer. (IP, ICMP, etc.)
- **Network Access / Link (TCP/IP)** ↔ OSI Data Link + Physical layers. (Protocols like Ethernet, Wi-Fi at the link layer.)

Thus: OSI's 1–2 map to "Network Access", OSI 3 = "Internet", OSI 4 = "Transport", OSI 5–7 = "Application" in TCP/IP terminology. (Some texts split TCP/IP into 5 layers including a separate "Network Interface" and "Physical" layer, but the general mapping is as above.)

3 (c) Compare IEEE 802.11a, b, g, n, ac standards.

Answer: These are successive Wi-Fi (802.11) standards, differing in frequency band, data rate, and features:

- **802.11a:** (1999) Uses 5 GHz band, OFDM modulation, up to 54 Mbps. Shorter range (higher frequency).
- **802.11b:** (1999) Uses 2.4 GHz, DSSS modulation, up to 11 Mbps. Longer range but lower speed.
- **802.11g:** (2003) Uses 2.4 GHz, OFDM (like 802.11a), up to 54 Mbps. Backward compatible with 802.11b devices.
- **802.11n:** (2009) Uses 2.4/5 GHz, MIMO (multiple antennas), wider channels (40 MHz), up to ~600 Mbps. Greatly increased throughput and range due to spatial multiplexing.
- **802.11ac:** (2013) Uses 5 GHz only, wider channels (80/160 MHz), higher-order modulation (256-QAM), multi-user MIMO, up to several Gbps (theoretical ~1.3 Gbps or more). It is backward-compatible with n.
- **802.11ax (Wi-Fi 6)** (emerging) and **802.11be (Wi-Fi 7)** further increase efficiency and speeds.

Each generation improves data rate and spectrum efficiency by using better modulation, more channels, or multiple antennas. For example, 802.11ac can exceed 1 Gbps using 80 MHz channels and MU-MIMO, while 802.11b was limited to 11 Mbps on 2.4 GHz.

4 (a) Define various error-control strategies: checksum, ARQ (sliding window), and forward error correction (FEC).

Answer:

- **Checksum:** A simple error-detecting code. A checksum is computed by summing blocks (words or bytes) of the data (usually modulo a fixed value) and sending this sum with the message ³⁰. The receiver recomputes the sum; if it differs, an error is detected. It catches many random errors but not as reliably as a CRC. For example, IP/TCP use a 16-bit checksum.
- **ARQ (Automatic Repeat reQuest) / Sliding Window:** An error-control method where the sender keeps a window of unacknowledged packets. Packets are numbered; the receiver sends ACKs for received frames. If an ACK is missing (timeout) or a frame is corrupted, the sender retransmits. Sliding window (like Go-Back-N or Selective Repeat) allows multiple frames in flight to improve throughput. It provides reliable delivery by retransmitting damaged or lost packets. Key features: sequence numbers, acknowledgments, retransmissions, and sliding window flow control.
- **FEC (Forward Error Correction):** Here the sender adds redundant bits (error-correcting codes) so the receiver can **correct** some errors without retransmission. Examples are Hamming codes, Reed–Solomon, convolutional codes with Viterbi decoding. For instance, a Reed–Solomon code on a block of data adds parity symbols; if some symbols are corrupted by noise, the receiver can recover the originals

using the code. FEC is used where retransmission is costly (satellite links, real-time video), at the expense of added overhead and complexity.

4 (b) Explain piggybacking of acknowledgments. In Stop-and-Wait, what happens if an ACK is lost?

Answer: Piggybacking: In bidirectional communication, a node that has data to send can include the ACK for the other side's data within its data frame, instead of sending a separate ACK frame. This saves bandwidth. For example, if Host A sends data to B, and B wants to send data back, B can carry A's ACK inside the header of its outgoing data packet.

ACK loss in Stop-and-Wait: If an ACK is lost (and the data packet actually arrived correctly), the sender will not hear the ACK before the timeout, so it will assume the packet or ACK was lost. It will retransmit the same data packet. The receiver, upon getting this duplicate packet, can detect that it is a duplicate (by sequence number) and discard it (since it already processed it). Thus, loss of ACK causes one extra transmission, but data is not corrupted because duplicates are identified and dropped.

4 (c) Compare 1-persistent, p-persistent, and non-persistent CSMA. Calculate slotted ALOHA throughput.

Answer:

- **1-persistent CSMA:** When a station has data, it senses the channel. If idle, it **immediately transmits** (with probability 1). If busy, it continuously senses until it becomes idle, then transmits right away. This is aggressive: it maximizes chance to send early but also high collision risk at idle transitions.

- **p-persistent CSMA:** Used in slotted channels. When the channel is idle, the station transmits with probability p in the next time slot; with probability $1 - p$, it defers to the next slot (again sensing). If the channel is busy, it defers. This randomizes access on slot boundaries, reducing collisions compared to 1-persistent.

- **Non-persistent CSMA:** A station senses the channel; if idle, it transmits immediately. If busy, it waits a random time (backoff) then senses again. This reduces collisions but can underutilize the channel (because stations wait even if the medium becomes free momentarily).

Slotted ALOHA throughput: As in 3(b) above, Slotted ALOHA with offered load G has throughput $S = Ge^{-G}$ and maximum of $1/e \approx 0.37$ when $G = 1$ ¹⁸. (About 36.8% of the slots carry a successful transmission at best.)

4 (d) Explain the operation of a transparent bridge.

Answer: A transparent bridge (or learning bridge) connects two LAN segments at the MAC level. It learns which MAC addresses are reachable on each port by inspecting the **source** address of incoming frames. It builds a forwarding table: MAC→port. When a frame arrives:

- It checks the destination MAC. If that address is known in its table, it forwards the frame out the corresponding port.
- If the destination is unknown, it **floods** the frame out all other ports (broadcast on the bridge).
- If the destination is on the same segment (incoming port), it drops the frame.

This process is “transparent” because hosts do not need to know the bridge is there – they send as usual. Over time, the bridge populates its table and typically only forwards frames where needed. Because it uses MAC-layer addresses, it works with any network protocol above. In essence, a transparent bridge learns network topology dynamically (no manual configuration) and only forwards when necessary, thus reducing unnecessary traffic across segments.

5 (a) What services does the network layer provide to the transport layer? (List three.)

Answer: The network layer typically offers the following service **types** to transport:

- **Unreliable datagram delivery:** Best-effort delivery of packets with no connection (e.g. IPv4). This is a connectionless service with no guarantees (some packets may be lost or arrive out of order).
- **Virtual-circuit service:** A connection-oriented service where a path is set up (e.g. MPLS/ATM virtual circuits) and all packets follow it in order. Transport sees a “connected” network service.
- **Broadcast/multicast delivery:** (If supported) the ability to send a single packet to multiple hosts (Broadcast for all hosts, Multicast for a group) at the network level. (Alternatively, one can list **reliable vs. unreliable** as separate services if using an advanced network layer with error control.)

Thus, the network layer can operate either connectionless or connection-oriented and may provide one-to-all or one-to-group delivery as needed.

5 (b) What services does the transport layer provide to the application layer? (List three, with examples.)

Answer: The transport layer provides end-to-end communication services to applications. Three examples are:

- **Connectionless datagram service:** (e.g. UDP) An application can send independent messages without establishing a connection. Service is simple and low-overhead but unreliable. *Example:* A DNS client sends a query in a single UDP message.
- **Reliable connection-oriented byte-stream service:** (e.g. TCP) An application can open a connection, then send a continuous stream of bytes reliably. The transport ensures in-order delivery and error recovery (via ACKs and retransmits). *Example:* HTTP uses TCP to reliably fetch web content.
- **Multiplexing/demultiplexing:** The transport layer uses port numbers to allow multiple applications to share a single host-to-host connection. It forwards incoming segments to the correct application based on port. *Example:* A web server and email server run on the same machine with different port numbers.

Additionally, transport may provide flow control or congestion control as part of reliability. In summary: (1) **Unreliable datagrams** (UDP) for simple queries, (2) **reliable byte streams** (TCP) for robust data transfer, and (3) **multiplexing** for addressing multiple apps (using ports) ³¹.

December 2022

1 (a) Why is bit stuffing needed and how is it done? Perform bit stuffing on the bit pattern 110001111111100001111100.

Answer: Bit stuffing is used in bit-oriented protocols (like HDLC) to ensure special bit patterns (like the flag 01111110) do not appear accidentally in data. After 5 consecutive ‘1’ bits in the data stream, a ‘0’ bit is **automatically inserted**. This guarantees that the flag pattern (six 1’s ended by 0) appears only as frame delimiters, not in the data ³². It also works on raw bit streams without regard to byte boundaries, making it more flexible than byte stuffing.

Applying **bit stuffing** to the pattern 110001111111100001111100: scan left-to-right and insert a 0 after any run of five 1’s. The original has runs of six 1’s twice. After each run of five 1’s, we stuff a 0:

```
Original: 110001 11111 1 0000 11111 00
Group runs:      ^^^^^^          ^^^^^ (stuff positions)
After stuffing: 110001 11111(0) 110000 11111(0) 00
```

So the stuffed sequence becomes 11000111111011000011111000. (We inserted a ‘0’ after each group of five 1’s.)

1 (b) Explain the concept of digital-to-digital encoding. Give an example.

Answer: Digital-to-digital encoding (line coding) is the process of converting a digital bitstream into a digital signal suitable for transmission on a physical medium ³³. Each bit (or group of bits) is mapped to a pattern of signal levels or pulses. For example, **NRZ-L (Non-Return-to-Zero-Level)** maps a '1' bit to a high voltage and '0' to a low voltage (or vice versa), holding the level for the full bit duration. **Manchester encoding** (also digital) represents each bit by a transition: a '0' might be a high-to-low transition and a '1' a low-to-high in the middle of the bit period. These encodings ensure synchronization and can include a DC component or not, depending on the scheme.

Example: The bit sequence 1011 might be sent in NRZ-L as voltage levels [+V, -V, +V, +V] (if we define '1'=+V, '0'=-V). In Manchester, 1011 might become [↓, ↑, ↑] transitions at each half-bit boundary (where ↓ means high-to-low, ↑ means low-to-high).

1 (c) What are the characteristics of a WAN? Compare client-server vs peer-to-peer networks.

Answer: A **Wide Area Network (WAN)** spans a large geographic area (cities, countries), often connecting multiple LANs. Characteristics: It typically uses leased telecommunications links or carrier networks. WANs may have lower data rates (kbps to Gbps) and higher latency than LANs. They require routers, public or private links, and often cost more per bit. WANs enable inter-organization connectivity (e.g. corporate campuses, internet backbones) ³⁴ ³⁵.

Client-Server vs. Peer-to-Peer: These are application-layer architectures (but relevant to networks). In **client-server**, one or a few nodes (servers) provide services, and clients request them. The roles are fixed: servers wait for requests, clients initiate. (Example: A web server responding to many browser clients.) In **peer-to-peer (P2P)**, every node can act as both client and server, sharing resources directly. No central server is required. (Example: BitTorrent file-sharing, or decentralized blockchain nodes.) P2P is more decentralized and can be robust, but client-server can centralize administration and is simpler for some tasks. In short, client-server has dedicated servers and passive clients ³⁶, whereas P2P treats all hosts equally ³⁷.

1 (d) Why is multiplexing important in networks? List basic multiplexing techniques.

Answer: Multiplexing allows multiple data streams to share the same physical medium, increasing efficiency. Instead of one user per link, multiplexing lets many signals coexist on one channel, making better use of bandwidth ³⁸ ³⁹. It is essential for combining data (voice, video, etc.) onto high-speed links, reducing infrastructure costs, and adapting to bursty traffic. For example, in TDM the link carries many telephone calls by interleaving voice samples in time slots.

Common multiplexing techniques include:

- **Time-Division Multiplexing (TDM):** Assigns each signal its own time slot in a repeating frame ⁴⁰. Each user transmits sequentially in turn (suitable for digital signals).
- **Frequency-Division Multiplexing (FDM):** Assigns each signal its own frequency band within the channel ⁴⁰. Each user modulates a carrier at a different frequency. (Example: cable TV uses FDM channels.)
- **Wavelength-Division Multiplexing (WDM):** In fiber optics, a form of FDM using different light wavelengths (colors) on one fiber.
- **Code-Division Multiplexing (CDM):** (e.g. CDMA) Assigns each signal a unique code to allow simultaneous transmission over the same frequency band.

These schemes allow multiple users or channels to share a single link.

2 (a) What is a checksum? Explain the key features of a sliding-window protocol.

Answer: A **checksum** is an error-detection code where the sender computes a small summary value from the data (e.g. summing all 16-bit words of a packet, then taking 1's complement). This checksum is sent with the packet. The receiver recomputes the sum; if the result is non-zero, an error is detected. Checksums are simple and catch many errors (as used in IP and UDP headers).

A **sliding-window** protocol is a flow-control/reliable transmission method where the sender can transmit multiple frames before waiting for ACKs. Features include:

- **Window of outstanding frames:** The sender maintains a window (size N) of unacknowledged frames. It can send new frames as long as the number in flight is below N .
- **Sequence numbers:** Frames are numbered; ACKs indicate the next expected sequence or the highest in-order frame received.
- **Cumulative ACKs:** The receiver typically ACKs the highest contiguous sequence received.
- **Retransmission on timeout:** If an ACK is not received in time for a frame, that frame (and possibly following frames, depending on variant) is retransmitted.
- **Flow control:** The receiver may advertise how many more frames it can buffer (sliding window also regulates send rate).

The sliding-window greatly improves link utilization by pipelining multiple frames ⁴¹. For example, TCP's window is a classic sliding-window: the sender can send up to the current congestion window, and each ACK (with window size) "slides" the window forward ⁴¹ ⁴².

2 (b) What is the main difference between static and dynamic routing? For IP address 197.22.1.0, what is the classful network address (Class A/B/C)?

Answer:

- **Static routing:** Routes are manually configured by a network administrator and do not change unless reconfigured. It has low overhead and high control, but it does not adapt to topology changes (if a link fails, static routes must be updated manually) ⁴³.
- **Dynamic routing:** Routers run algorithms (RIP, OSPF, etc.) that automatically share and update routes. They adapt to network changes (a failed link triggers route recomputation) ⁴³. Dynamic routing requires more CPU and bandwidth (for routing updates) but is more fault-tolerant and scalable.

For the address **197.22.1.0**: The first octet 197 falls between 192 and 223, so it is a **Class C** address ⁴⁴. The classful network address (with default mask 255.255.255.0) is **197.22.1.0** itself (network ID), and the range would be 197.22.1.0–197.22.1.255. (It is *not* Class A or B, so Class C applies.)

3 (a) Calculate the end-to-end delay for a three-link path. Each link's length and one-way propagation speed: 600 km, 1200 km, 300 km at 2×10^6 m/s. The packet length is 10,000 bits plus a 100-bit header (total 10,100 bits), and each link's transmission rate is 500 kb/s. Each router adds 2 ms processing per packet.

Answer: Total end-to-end delay = sum of all propagation delays + all transmission delays + all processing delays. Compute each:

- **Transmission delay per link:** Packet size = 10,100 bits. Rate = 500,000 b/s. $T_{tx} = 10,100/500,000 = 0.0202$ s per link. For 3 links: $3 \times 0.0202 = 0.0606$ s.
- **Propagation delay per link:** $T_{prop} = d/v$. Link1: $600,000/2 \times 10^6 = 0.30$ s. Link2: $1,200,000/2 \times 10^6 = 0.60$ s. Link3: $300,000/2 \times 10^6 = 0.15$ s. Sum = 1.05 s.
- **Processing delay:** 3 routers $\times$ 2 ms each = 6 ms = 0.006 s.

Total = 1.05 + 0.0606 + 0.006 $\approx$ 1.117 seconds.

Using the formulas $T_{tx} = L/R$ and $T_{prop} = d/v$ ³ ⁴, the total end-to-end delay is about **1.12 s**.

3 (b) Explain the count-to-infinity problem in distance-vector routing and how RIP avoids it.

Answer: In distance-vector routing, count-to-infinity occurs when a network goes down or a link breaks: routers increment the metric for that destination in steps (1,2,3,...) each update cycle, potentially looping forever if not corrected. RIP limits this with “infinite” = 16 hops. RIP **avoids loops** via:

- **Split Horizon with Poison Reverse:** A router does not advertise a route back on the interface from which it learned it (or advertises it with infinite metric), preventing two routers from falsely telling each other a route is still valid ⁴⁵.
- **Route Poisoning:** When a route fails, RIP immediately advertises it with metric 16 (infinity) to force all routers to drop it ⁴⁶.
- **Hold-Down Timers:** If a route becomes unreachable, routers ignore any changes to that route for a period to allow stable convergence.

In RIP specifically, once a neighbor signals the route is unreachable, the metric is set to 16 (unreachable). Because of these mechanisms (split horizon and poisoned updates), RIP quickly “kills” the bad route instead of letting the count slowly increase, preventing endless loops ⁴⁵.

3 (c) Differentiate congestion control vs. flow control (briefly).

Answer: (Repeat from 1(d) with context) Flow control prevents the sender from overwhelming the receiver's buffer (end-to-end). It regulates the sender's rate to the receiver's capacity. Congestion control prevents network overload by regulating the total traffic entering the network (even if the receiver is fine). Flow control concerns receiver speed, while congestion control concerns network capacity.

4 (a) List the network-layer services provided to the transport layer (at least three types).

Answer: The network layer can offer:

- **Connectionless (datagram) service:** Sends packets without prior setup, independent delivery of each packet (like IPv4).
- **Connection-oriented (virtual-circuit) service:** Requires setup of a path (like X.25 or MPLS VC) before sending, then in-order delivery along that path.
- **Unicast, broadcast, multicast addressing:** (service modes) Unicast for one-to-one, broadcast (on local LANs) for one-to-all, and multicast (IPv4/6) for one-to-group. These define how packets are delivered to hosts.

Some sources list reliable vs unreliable, or guaranteed delivery vs best-effort, but in a basic IP-based network, the main modes are unicast datagrams (connectionless) or virtual-circuit (connection-oriented) service.

4 (b) List the transport-layer services provided to the application layer. Give examples.

Answer: The transport layer typically provides:

- **Reliability / Connection-oriented byte-stream service:** (TCP) Ensures data is delivered error-free and in order. *Example:* Web (HTTP), file transfer (FTP) use TCP so the application sees a reliable stream.
- **Unreliable datagram service:** (UDP) Provides simple, low-overhead delivery with no connection setup and no guarantees. *Example:* DNS or DHCP, where occasional packet loss is tolerable.
- **Multiplexing/demultiplexing:** Using port numbers to allow multiple applications on one host to communicate. *Example:* A machine runs both a web server (port 80) and email server (port 25), and the transport layer directs each received segment to the correct app.

Additional services include flow control (prevent receiver overflow) and congestion control (avoid network overload) for connection-oriented transport.

5 (a) What is the format and use of digital signatures? Give an example.

Answer: A **digital signature** ensures message integrity, authentication, and non-repudiation. To sign a message, the sender typically: (1) hashes the message with a cryptographic hash (MD5/SHA-256), (2) encrypts the hash with their private key (forming the signature). The recipient decrypts it with the sender's public key and compares it to a hash of the received message; if they match, the sender is authenticated and the message hasn't changed. Digital signatures are analogous to handwritten signatures but are encrypted and tied to the exact message.

Example: In email (S/MIME or PGP), a sender signs an email by hashing and encrypting the hash with their RSA private key. The recipient uses the sender's public key to verify.

5 (b) Explain what a proxy server is. What security risks does it introduce?

Answer: A **proxy server** is an intermediary that forwards requests from clients to other servers (often on the Internet). Clients send their requests to the proxy (e.g. HTTP or SOCKS proxy); the proxy then makes the actual network request and returns results to the client. Proxies can provide caching, content filtering, or anonymity.

Security risks include:

- **Privacy:** The proxy can see all your traffic (URLs, data) and possibly log or alter it.
- **Trust:** Users must trust the proxy not to tamper with data. A malicious proxy could inject malware or filter responses.
- **Authentication:** If not configured securely, proxy credentials might be intercepted.
- **Single point of failure:** If a proxy is compromised, it can affect all clients using it.

In summary, proxies break the end-to-end model: data is decrypted at the proxy. If the proxy is not fully trusted, it becomes a vulnerability in the path.

June 2023

1 (a) Quantize an analog signal ranging from -5 to $+3$ volts into 16 equal levels. Compute the step size and quantization intervals. What is the quantized value of -3.5 V?

Answer: The signal range is 8 volts (from -5 to $+3$). With 16 levels, the quantization step size is $\Delta = 8/16 = 0.5$ V per level. The levels (midpoints) might be at $-4.75, -4.25, -3.75, -3.25, \dots, +2.75$.

Divide the range into intervals of width 0.5 V: e.g. $[-5.00, -4.50), [-4.50, -4.00), [-4.00, -3.50), [-3.50, -3.00), \dots$. The sample -3.5 V lies at the boundary between $[-4.00, -3.50)$ and $[-3.50, -3.00)$. Typically, -3.50 would be quantized to the midpoint of one of these intervals; if we assume mid-rise, -3.5 maps to -3.75 (lower midpoint) or -3.25 (upper midpoint) depending on convention. Usually, one would choose $-3.50 \rightarrow -3.50$ by symmetry (or to nearest midpoint). For simplicity, say the quantized value is **-3.50 V** (no change), or if forced to midpoints, $-3.50 \rightarrow -3.75$ V. In any case, the step is **0.5 V**.

1 (b) What is the minimum and maximum length of an IEEE 802.3 (Ethernet) frame? Give the differences between 10Base2 (coax) and 10Base-T (UTP).

Answer: An IEEE 802.3 Ethernet frame has a **minimum size** of 64 bytes (octets) including header and CRC, and a **maximum size** of 1518 bytes (standard Ethernet frame without VLAN tag) ³. (With the 4-byte FCS, that is 64–1518 bytes in total, or 68–1522 if including VLAN.)

10Base-2 (Thin Ethernet coax) vs 10Base-T (Twisted-Pair):

- 10Base-2 uses coaxial cable in a bus topology. Stations tap into the coax; the cable must be terminated at both ends with 50Ω terminators. It supports about 185-meter maximum length per segment, and up to ~30 nodes per segment (BNC T-connectors). No active hub is needed; the coax itself is the medium.
- 10Base-T uses unshielded twisted-pair (UTP) cable in a star topology. Each device plugs into a central hub/switch. Segment length is up to 100 meters per node. 10Base-T is easier to install and reconfigure (just add cables to the hub). A hub (or switch) is required. Failure of the hub will down the network, whereas in 10Base-2, a single device removal can break the chain if not properly re-terminated.

In summary, 10Base-2 is a shared bus (coax), while 10Base-T is a star (UTP) – both operate at 10 Mbps but use different cabling and topologies.

1 (c) List and explain congestion avoidance (non-adaptive, open-loop) and congestion control (adaptive, closed-loop) mechanisms.

Answer: To avoid or control congestion in networks, two general approaches exist:

- **Open-loop (Preventive) Policies:** These adjust traffic before congestion happens. For example, **flow control/traffic shaping** (e.g. leaky bucket) regulates the rate at which packets enter the network to smooth bursts. **Admission control** in networks like ATM can refuse new calls if the network is “full” or allocate bandwidth tokens. These are non-adaptive (they do not rely on feedback); they set rates based on contracts or limits. *Example:* A token bucket limits an application’s send rate to a predetermined average, preventing sudden overloads.
- **Closed-loop (Reactive) Control:** These detect congestion and then respond. For example, **TCP’s congestion control** is a closed-loop algorithm: when packet loss or delay indicates congestion, TCP reduces its sending window (e.g. slow-start, AIMD). **Router queue management** like Random Early Detection (RED) proactively drops or marks packets when queues start filling, signaling senders to slow down. **Explicit feedback schemes** (like ICMP Source Quench, or Explicit Congestion Notification) also tell the sender to lower rate. Closed-loop methods monitor network state (loss, delay, queue length) and adapt.

In summary: *Congestion avoidance* techniques set traffic limits in advance (shaping, policing), while *congestion control* reacts to actual congestion signals (packet drops, timeouts) by slowing senders (e.g. TCP backoff). Both aim to keep the network out of overload.

1 (d) What is public-key cryptography? Explain RSA key generation (with example).

Answer: Public-key cryptography (asymmetric cryptography) uses a pair of keys: a **public key**, which anyone may know, and a **private key**, kept secret. A message encrypted with the public key can only be decrypted by the corresponding private key, enabling secure exchange without a shared secret ⁴⁷. Key pairs are mathematically linked (e.g. one-way functions like multiplication of large primes).

RSA key generation: As described earlier in 1(f). In summary:

1. Choose two primes p, q , compute $n = pq$ and $\phi = (p - 1)(q - 1)$.
 2. Pick public exponent e such that $\gcd(e, \phi) = 1$.
 3. Compute $d = e^{-1} \pmod{\phi}$ (the private exponent).
- Public key = (n, e) , Private key = (n, d) ⁸.

Example: Let $p = 7, q = 11$. Then $n = 77$, $\phi = 60$. Choose $e = 17$. Solve $17d \equiv 1 \pmod{60}$, yielding $d = 53$. Public key = $(77, 17)$; private = $(77, 53)$, as before.

1 (e) Define virus, worm, and Trojan horse. What are their differences?

Answer: These are types of malware with distinct behaviors ⁴⁸ :

- **Virus:** A program that attaches itself to other executable files. It replicates by infecting hosts when the host program runs. It cannot spread on its own (a user must run the infected file) ⁴⁸ . When executed, it may corrupt data or replicate to other files.
- **Worm:** A standalone program that self-replicates over networks. It does **not** need to attach to a host file. Worms exploit network connections to spread to other machines automatically ⁴⁸ . They often consume bandwidth/resources by copying themselves.
- **Trojan Horse:** A malicious program disguised as legitimate software. Unlike viruses and worms, Trojans do **not** self-replicate. They rely on deceiving the user to install/run them (e.g. pretending to be a useful app). Once installed, a Trojan can steal data or open backdoors ⁴⁸ .

In summary (table):

- Replication: *Virus* replicates by attaching to files; *Worm* replicates on its own; *Trojan* does not replicate.
- Activation: *Virus/Worm* execute when launched; *Trojan* runs as a hidden malicious payload.
- Self-contained: Worms are independent; viruses depend on host files; Trojans depend on users installing them.

1 (f) Name four types of noise in communication links: thermal, intermodulation, crosstalk, impulse. Describe each.

Answer:

- **Thermal noise (Johnson noise):** Caused by the random motion of electrons in resistive elements. It is present in all electrical conductors and has a wide (white) spectrum ⁴⁹ . It increases with temperature and bandwidth (e.g. kTB). Thermal noise adds a baseline uncertainty to the signal.
- **Intermodulation noise:** Occurs when multiple signals share a non-linear medium or device. Two signals at frequencies f_1 and f_2 can mix to produce spurious frequencies (sums/differences $f_1 \pm f_2$) that interfere with other signals ⁵⁰ . For example, if two channels exist on the same cable, non-linearities can create cross-terms.
- **Crosstalk:** Interference from adjacent channels. When two transmission lines are near each other, a signal on one can induce noise in the other (like hearing nearby conversations). In cables, it manifests as one pair coupling into another. Crosstalk can be near-end (NEXT) or far-end (FEXT) depending on where it's measured ⁵¹ .
- **Impulse noise:** Irregular, high-amplitude spikes of short duration caused by external events (e.g. lightning, electrical switching) ⁵² . Impulse noise is a major source of bit errors in digital signals, since one impulse can corrupt multiple bits. Examples include switching transients or automotive ignition noise.

Each of these impairs the transmitted signal in different ways: thermal adds a constant background noise, intermodulation and crosstalk add interfering signals, and impulses cause sporadic large errors.

2 (a) What is the difference between transmission delay and propagation delay?

Answer: Repeat for clarity: Transmission delay is the time to put all packet bits onto the link (bit-rate dependent): $T_t = \frac{\text{packet length (bits)}}{\text{link bandwidth}}$ ³ . Propagation delay is the time for a bit to travel through the medium: $T_p = \frac{\text{distance}}{\text{propagation speed}}$ ⁴ . Transmission depends on packet size and data rate; propagation depends on physical distance and signal velocity.

2 (b) What is PCM and why is PAM used in it?

Answer: Pulse Code Modulation (PCM) is a method for digitizing analog signals: the analog waveform is sampled and each sample is quantized to a binary code. Before quantization, the analog waveform is sampled by *pulse amplitude modulation (PAM)* – i.e., the amplitude of the waveform is captured at regular

intervals and turned into a train of pulses whose amplitudes follow the signal. In other words, PAM is the stage that grabs each analog sample and converts it to a pulse; PCM then encodes that pulse amplitude into bits. Thus PAM is needed in PCM to create discrete pulse samples from the continuous signal. Without PAM sampling, PCM wouldn't have values to quantize.

2 (c) Compare circuit switching vs packet switching.

Answer: (Same as 1(c) above) **Circuit switching** sets up a fixed path before communication (like a dedicated circuit). **Packet switching** breaks data into packets that travel independently with no reserved path. Example of circuit: legacy telephone. Example of packet: Internet (IP). In circuit switching, once the call is set up, bandwidth is exclusively reserved (so guaranteed transmission but possible idle waste). In packet switching, link capacity is shared dynamically (so flexible but variable delay and possible contention).

2 (d) What is the main function of each OSI layer?

Answer: Briefly, the 7 OSI layers do:

1. **Physical:** Bit-level transmission over the physical medium (cables, radio). Defines electrical/optical signals, voltages, connectors.
2. **Data Link:** Frames bits into frames, provides physical addressing (MAC), error detection (CRC), and possibly error recovery/flow control on a single link.
3. **Network:** Performs host-to-host packet delivery across networks. Handles logical addressing (IP), routing, and fragmentation.
4. **Transport:** End-to-end communication between hosts. Provides segmentation/reassembly, reliability (if needed), flow control, and multiplexing (ports). E.g. TCP/UDP.
5. **Session:** Manages sessions/connection dialogs between applications (e.g. sets up, coordinates, and tears down sessions). Handles things like dialogue control, synchronization (e.g., checkpoints).
6. **Presentation:** Translates data formats (encryption, compression, character encoding). Ensures that data is presented in a usable form (e.g. converts from EBCDIC to ASCII or encrypts/decrypts).
7. **Application:** The interface for user applications. Provides network services to end-user processes (e.g. HTTP, SMTP, FTP).

In practice, layers 5–7 are often combined into one “application layer” in the TCP/IP model, but OSI defines them separately for conceptual clarity.

3 (a) Describe the CRC error detection method.

Answer: CRC (Cyclic Redundancy Check) treats a message as a binary polynomial and divides it by a fixed generator polynomial. The remainder of this binary division (of degree < generator degree) is appended to the message as the CRC bits ¹. The receiver performs the same polynomial division on the received message+CRC; if the remainder is nonzero, an error is detected. CRCs are good at detecting burst errors. The generator polynomial is agreed by sender and receiver. Example: for generator $x^3 + x + 1$ (1001), if the received message including CRC is not divisible by 1001, an error occurred.

3 (b) What is piggybacking in ARQ, and how is a lost ACK handled in Stop-and-Wait?

Answer: Piggybacking was covered above. In Stop-and-Wait ARQ, if an ACK is lost, the sender's timeout expires and it **retransmits** the same frame. The receiver will see the duplicate frame (based on sequence number) and discard it (since it already got it the first time). Thus a lost ACK results in one extra transmission, but the data is not corrupted (duplicate dropped by the receiver). The protocol remains reliable by retransmissions upon timeout.

3 (c) Compare 1-persistent, p-persistent, and non-persistent CSMA. What is the maximum throughput of Slotted ALOHA?

Answer: (Same as 4(c) above.) 1-persistent: transmit immediately when idle, keep sensing if busy. p-persistent: in slotted channel, transmit with prob p on an idle slot, else wait. Non-persistent: if busy, wait a random time before retry. Slotted ALOHA's max throughput is $1/e \approx 0.37$ of the channel capacity ¹⁸.

3 (d) What is a transparent bridge and how does it operate?

Answer: (Same as 3 (d) Dec.) A transparent bridge forwards Ethernet frames between LAN segments by learning MAC addresses. It is "transparent" to hosts (hosts send normally). The bridge learns by listening to incoming frames: it records each source MAC and the port it came from. For each frame: if the destination MAC is in its table, it forwards only on the associated port; if unknown, it floods the frame to all other ports. This way, it connects two Ethernet segments without requiring any configuration, and only forwards frames as needed.

4 (a) State the services provided by the network layer to the transport layer (3 types).

Answer: (Similar to 5(a) Dec.) The network layer typically offers:

1. **Unreliable datagram delivery:** Best-effort packet delivery (connectionless).
2. **Virtual circuit service:** Connection-oriented delivery (pre-set path, reliable ordering).
3. **Broadcast/Multicast:** Ability to deliver a packet to all hosts (broadcast on LAN) or group (multicast addresses).

These cover the common modes: unicast datagrams, connection paths, and group delivery options.

4 (b) State the services provided by the transport layer to the application layer (3 types).

Answer: (As 5(b) Dec.) The transport layer provides:

1. **Reliable byte-stream (connection-oriented):** e.g. TCP. Example: file transfer.
2. **Unreliable datagrams (connectionless):** e.g. UDP. Example: streaming audio.
3. **Multiplexing/demultiplexing:** using port numbers to allow multiple apps. Example: web server and DNS server on same host.

4 (c) Explain the modulo function in cryptography. Outline elliptic-curve cryptography (ECC) basics.

Answer: The **modulo function** computes the remainder of division. For integers a and modulus m , $a \bmod m$ is the remainder when a is divided by m . For example, $10 \bmod 3 = 1$. Modular arithmetic is fundamental in cryptography (e.g. RSA uses exponentiation mod n).

Elliptic-Curve Cryptography (ECC): ECC is an asymmetric scheme based on the algebraic structure of elliptic curves over finite fields ⁵³. An elliptic curve has an equation like $y^2 = x^3 + ax + b \pmod{p}$. A public key is a point on this curve (derived from a private scalar and a base point). The security relies on the elliptic curve discrete logarithm problem: given P and kP , it's hard to find k . ECC achieves similar security with smaller key sizes than RSA. For example, a user chooses private key k (a large integer), computes public key $Q = kG$ where G is a known base point on the curve. To encrypt, one uses the recipient's public point Q and a random ephemeral point; to sign, one performs arithmetic on the curve. ECC is now widely used (e.g. in secure messaging and TLS) because of its efficiency ⁵³.

5 (a) What is a vulnerability? Give two examples of software vulnerabilities.

Answer: A **vulnerability** is a flaw or weakness in a system or software that could be exploited to compromise security. For example:

- **Buffer overflow:** When a program does not properly check input size, an attacker can overflow memory and execute malicious code.
- **SQL injection flaw:** A web app that directly includes user input in SQL queries can allow attackers to

manipulate the database.

Other examples: Cross-site scripting (XSS), improper authentication checks, unpatched bugs. Vulnerabilities can lead to unauthorized access, data breaches, or malware infection.

5 (b) Why is a DNS server called a “root” in its context? What is a TLD and give an example.

Answer: In DNS, the **root server** is the top of the DNS hierarchy. It is called “root” because it holds the root zone (denoted “.”) and knows how to direct queries for all top-level domains. Any DNS lookup ultimately traces up to a root server to resolve unknown domains. The root is akin to “/” in a filesystem.

A **TLD (Top-Level Domain)** is the last segment of a domain name, indicating a highest level of the DNS hierarchy. Examples: `.com`, `.org`, `.uk`, `.jp`. For instance, in “example.com”, “.com” is the TLD. Root servers know about TLD name servers, which then direct to second-level domains.

5 (c) Explain NAT (Network Address Translation) and mention a limitation.

Answer: NAT is a technique that allows many devices on a private LAN (with private IPs) to share a single public IP address. A NAT router rewrites the source IP and port of outgoing packets (to the router’s public IP) and keeps a translation table. Incoming replies are mapped back to the correct internal host by reversing this process. NAT conserves IPv4 addresses and adds a layer of obscurity.

Limitation: NAT breaks end-to-end addressability. Inbound connections must be explicitly forwarded (port forwarding), and protocols that embed IP addresses in payload can fail unless application-level gateways are used. Also, NAT complicates peer-to-peer connections and some VPNs, since the true IPs are hidden. It makes some applications (like VoIP, direct games) harder without special handling.

5 (d) Describe RMI (Remote Method Invocation) in brief. Why is it used?

Answer: RMI is a Java API that allows an object on one Java Virtual Machine to invoke methods on an object in another JVM (possibly on a different host) as if it were local. It handles network communication, object serialization, and stub/skeleton plumbing so the programmer calls remote methods directly. RMI is used for building distributed applications in Java, enabling remote procedure calls in an object-oriented manner. It simplifies invoking functionality on a server from a client (for example, querying a remote database through method calls).

5 (e) What is a Digital Certificate (X.509)? What fields does it typically contain?

Answer: An X.509 digital certificate is a signed credential that binds an entity’s identity to a public key. It is issued by a Certificate Authority (CA) after verifying the subject’s identity. Key fields in an X.509 certificate include: Subject name (the identity, e.g. a server’s domain), Issuer name (CA’s identity), Subject’s public key, validity period (notBefore/notAfter dates), serial number, and CA’s digital signature. Optionally it contains extensions like key usage. Applications use these certificates (e.g. in TLS) to verify that a public key truly belongs to the claimed subject.

5 (f) Explain ARQ (Automatic Repeat reQuest) and Forward Error Correction.

Answer: ARQ (Automatic Repeat reQuest) is an error-control method where the receiver detects errors (e.g. via checksum/CRC) and requests retransmission. In practice, corrupted or lost frames are resent. Examples include Stop-and-Wait ARQ and Sliding Window ARQ (like TCP). ARQ provides reliability by retransmission.

Forward Error Correction (FEC) adds redundant data to the message so that the receiver can **correct** some errors without asking for retransmission. Examples: Hamming code, Reed–Solomon code in CDs, convolutional codes. The transmitter encodes k data bits into $n > k$ code bits; the receiver, upon getting a possibly corrupted codeword, can use the code’s error-correcting properties to recover the

original data (up to the code's error-correcting capability). FEC is used when retransmissions are costly or impossible (e.g. deep-space links, streaming video).

5 (g) Discuss possible security threats caused by code modification in software.

Answer: Code modification threats include **malicious code injection** or unauthorized patching. For example, if an attacker adds or alters code in an application or OS, they could:

- Introduce malware (trojan) that steals data or opens backdoors.
- Bypass authentication checks to gain unauthorized access.
- Sabotage functionality (destroy data, cause crashes).

Code tampering breaks the integrity of software. It can happen through vulnerabilities that allow writing to executable memory, or by replacing binaries on disk. Protecting against such threats requires code signing, checksums, secure boot, and strict access controls.

5 (h) How can a host behind NAT scan the IPv6 address of LAN devices?

Answer: Unlike IPv4, NAT is generally not used with IPv6 (each device can have a public IPv6 address). A host can scan LAN IPv6 addresses directly if it knows the subnet. For example, using `nmap -6` on the LAN's IPv6 prefix can find active hosts. Alternatively, the host can use ICMPv6 "Neighbor Discovery" (ping all addresses, use multicast "all-nodes" addresses) to discover IPv6 devices on the same link. Since NAT typically is IPv4-only, scanning over IPv6 is straightforward by iterating through the global/unicast range assigned to the network. (In summary, IPv6's abundant address space and lack of NAT means hosts have routable IPv6 addresses, so one can directly contact or ping-sweep them. There is no hidden NAT for IPv6 as in IPv4.)

Sources: Authoritative network and cryptography references and textbooks. All specific definitions and formulas were taken from the cited resources above

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